

# Product Requirements Document (PRD)

## **DOCUMENT INFORMATION**

| Field        | Value                  |
|--------------|------------------------|
| Product Name | Cryptocurrency Tracker |
| Version      | 1.0                    |
| Author       | Krishna                |
| Date         | July                   |
| Stakeholder  | End Users/Developer    |
| Status       | Draft                  |

## **EXECUTIVE SUMMARY**

### **Problem Statement**

Prices for cryptocurrencies fluctuate quickly throughout the market, making it challenging for users to keep an eye on several digital assets, see trends, and make defensible decisions utilizing dispersed data from several sources.

### **Proposed Solution**

In order to make market monitoring and research easier, Cryptocurrency Tracker is a web-based program that offers real-time cryptocurrency prices, market statistics, historical charts, portfolio tracking, and customized watchlists on a single, user-friendly platform.

### **Statement of Vision**

To provide a dependable, scalable, and intelligent cryptocurrency tracking system that enables users to make well-informed investment decisions by providing them with real-time market insights, sophisticated analytics, and easy portfolio management.

## **BUSINESS OBJECTIVES**

### **Goal 1:**

Provide real-time cryptocurrency market data through a fast, reliable, and user-friendly platform.

### **Goal 2:**

Enables users to monitor cryptocurrency prices, market trends, and personal portfolios from a single dashboard.

### **Goal 3:**

Deliver accurate market insights and analytical tools to support informed investment decisions.

## **Success Metrics (KPIs)**

| KPI                                | Target                                                         |
|------------------------------------|----------------------------------------------------------------|
| User Registration Rate             | ≥ registered users within the first 3 months                   |
| Daily Active Users (DAU)           | ≥ 50 active users per day                                      |
| Project Completion Rate            | 100% of planned MVP features completed before final submission |
| User Retention                     | ≥ 60% of users return within 30 days                           |
| Customer Satisfaction Score (CSAT) | ≥ 4.5/5 average user rating                                    |

## **TARGET AUDIENCE**

### **Primary Users**

The primary users of the Cryptocurrency Tracker are cryptocurrency investors, traders, beginners, and market enthusiasts who want to monitor real-time prices, analyze market trends, manages their portfolios, and make informed investment decisions through a single platform.

## **User Persona 1**

**Name:** Rahul Sharma

**Age:** 28

**Occupation:** Software Engineer

### **Goals:**

- Track real-time cryptocurrency prices.
- Monitor investment portfolio performance.
- Receive market insights before making investment decisions.
- Save favorite cryptocurrencies in a watchlist.

### **Pain Points:**

- Switching between multiple websites for market information.
- Difficulty tracking multiple cryptocurrencies simultaneously.
- Lack of a personalized dashboard.
- Time-consuming market analysis.

## **User Persona 2**

**Name:** Krishna

**Age:** 22

**Occupation:** College Student

### **Goals:**

- Learn about cryptocurrencies.
- Track trending coins.
- Understand historical price movements.
- Practice investing without using multiple platforms.

### **Pain Points:**

- Limited knowledge of cryptocurrency markets.
- Confusing interfaces on professional trading platforms.
- Difficulty understanding market trends.
- Information scattered across different websites.

## **USER STORIES**

### **1. View Live Cryptocurrency Prices**

As a user, I want to view real-time cryptocurrency prices so that I can stay updated with the latest market movements.

### **2. Search for Cryptocurrencies**

As a user, I want to search for a specific cryptocurrency so that I can quickly find detailed information about it.

### **3. View Coin Details**

As a user, I want to view detailed information about a cryptocurrency so that I can analyze its market performance before investing.

### **4. View Historical Price Charts**

As a user, I want to view historical price charts so that I can understand price trends over different time periods.

### **5. View Market Statics**

As a user, I want to view global cryptocurrency market statistics so that I can understand the overall market condition.

### **6. Receive Price Alerts**

As a registered user, I want to receive price alerts so that I can react quickly when a cryptocurrency reaches my target price.

### **7. Read Market News**

As a user, I want to read the latest cryptocurrency news so that I can stay informed about important market events.

## **SCOPE**

### **Core Features**

- User Registration and Authentication
- Secure Login and Logout
- Real-Time Cryptocurrency Price Tracking
- Cryptocurrency Search and Filtering
- Dashboard with Market Overview
- Detailed Coin Information
- Historical Price Charts
- Portfolio Management
- Watchlist Management
- Global Market Statistics

- Responsive Web Interface

### **Out of Scope (Version 1.0)**

- Buying/Selling
- Payment Gateway Integration
- Mobile Application (Android/iOS)
- Blockchain Wallet Integration
- AI-Based Price Prediction
- Social Media or Community Features
- Multi-Language Support
- Mining Features
- NFT Marketplace Integration

## **FUNCTIONAL REQUIREMENTS**

### **Module 1: Authentication**

#### **Features**

User Authentication

#### **Description**

Users should be able to securely register, log in, and manage their accounts.

#### **Requirements**

| ID     | Requirements                                                   |
|--------|----------------------------------------------------------------|
| FR-001 | Users can create a new account using their email and password. |
| FR-002 | Registered users can securely log in to the application.       |
| FR-003 | Users can reset their password using their registered email.   |

### **Module 2: Dashboard**

#### **Features**

Market Dashboard

## Description

The dashboard provides an overview of the cryptocurrency market with real-time data.

## Requirements

| ID     | Requirements                                                      |
|--------|-------------------------------------------------------------------|
| FR-005 | Display real-time cryptocurrency prices.                          |
| FR-006 | Display market capitalization, trading volume, and price changes. |
| FR-007 | Show top gainers and losers.                                      |

## Module 3: Cryptocurrency Search

### Feature

Search and Filter

### Description

Users can quickly search and filter cryptocurrencies.

### Requirements

| ID     | Requirement                                                                         |
|--------|-------------------------------------------------------------------------------------|
| FR-009 | Users can search cryptocurrencies by name or symbol.                                |
| FR-010 | Users can filter cryptocurrencies by market capitalization, price, and performance. |
| FR-011 | Search results should update dynamically.                                           |

## Module 4: Coin Details

### Features

Cryptocurrency Details

### Description

Users can view detailed information about any cryptocurrency.

## Requirements

| ID     | Requirements                                                                          |
|--------|---------------------------------------------------------------------------------------|
| FR-012 | Display current price, market capitalization, trading volume, and circulating supply. |
| FR-013 | Display historical price charts for multiple time ranges (24H, 7D, 30D, 1Y).          |
| FR-014 | Display percentage price changes over different periods.                              |

## Module 5: Portfolio Management

### Features

Portfolio Tracking

### Description

Registered users can manage and track their cryptocurrency investments.

### Requirements

| ID     | Requirements                                       |
|--------|----------------------------------------------------|
| FR-015 | Users can add cryptocurrencies to their portfolio. |
| FR-016 | Users can edit or remove portfolio entries.        |
| FR-017 | Display total portfolio value and profit/loss.     |

## Module 6: Watchlist

### Feature

Watchlist Management

### Description

Users can save their favorite cryptocurrencies for quick access.

## Requirements

| ID     | Requirements                                              |
|--------|-----------------------------------------------------------|
| FR-019 | Users can add cryptocurrencies to a watchlist.            |
| FR-020 | Users can remove cryptocurrencies from the watchlist.     |
| FR-021 | Watchlist should be synchronized with the user's account. |

## Module 7: User Profile

### Features

Profile Management

### Description

Users can manage their personal account information.

### Requirements

| ID     | Requirements                                |
|--------|---------------------------------------------|
| FR-022 | Users can update their profile information. |
| FR-023 | Users can change their password securely.   |
| FR-024 | Users can delete their account if desired.  |

## NON-FUNCTIONAL REQUIREMENTS

### 1. Performance

- The application should load the dashboard within **3 seconds** under normal network conditions.
- API responses should be returned within **2 seconds** for most requests.
- The system should efficiently handle **at least 100 concurrent users**.
- Cryptocurrency data should refresh automatically at configurable intervals.



## 2. Security

- User authentication shall be implemented using **JWT (JSON Web Tokens)**.
- All passwords shall be securely stored using **bcrypt hashing**.
- All communication between the client and server shall use **HTTPS**.
- Protected API endpoints shall require valid authentication tokens.
- User input shall be validated to prevent common security vulnerabilities.

## 3. Scalability

- The application should support deployment on cloud platforms.
- The backend architecture should allow horizontal scaling when user traffic increases.
- The database should support future migration from SQLite to PostgreSQL if required.

## 4. Reliability

- The application should maintain an uptime target of **99%** during deployment.
- The system should handle API failures gracefully by displaying meaningful error messages.
- The application should recover from temporary network failures without data loss.

## 5. Usability

- The interface should be intuitive and easy for both beginners and experienced cryptocurrency users.
- The application should have a responsive design for desktop, tablet, and mobile devices.
- Navigation between pages should be simple and consistent.

## 6. Accessibility

- The application should follow basic **WCAG 2.1 Level AA** accessibility guidelines where practical.
- Text should have sufficient color contrast for readability.
- Interactive elements should be accessible using keyboard navigation.

- Images and icons should include descriptive alternative text where applicable.

## 7. Compatibility

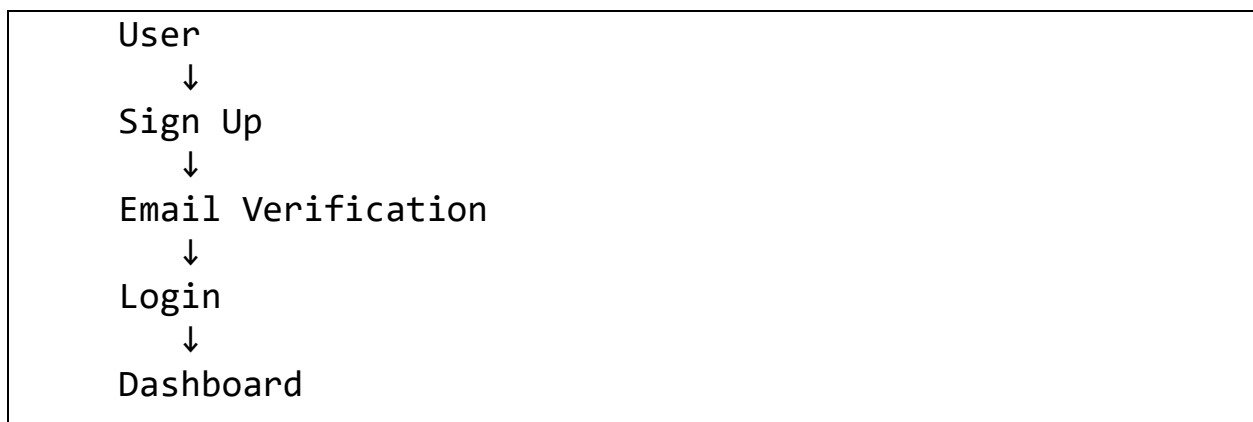
- The application should support the latest versions of major web browsers, including:
  - Google Chrome
  - Mozilla Firefox
  - Microsoft Edge
  - Safari

## 8. Maintainability

- The codebase should follow a modular architecture.
- Source code should be properly documented and version-controlled using Git.
- The application should be designed to allow new features to be added with minimal changes to existing code.

## USER FLOW

### 1. Registration Flow



## 2. Login Flow

User

↓

Login

↓

Authentication

↓

Dashboard

## 3. Cryptocurrency Search Flow

User

↓

Search Cryptocurrency

↓

View Search Results

↓

Select Cryptocurrency

↓

Coin Details Page

## 4. Portfolio Management Flow

User

↓

Portfolio

↓

Add / Edit Holdings

↓

Save Portfolio

↓

View Portfolio Summary

## 5. Watchlist Flow

User

↓

Browse Cryptocurrency

↓

Add to Watchlist

↓

Watchlist Updated

↓

View Watchlist

## 6. Market Analysis Flow

User

↓

Select Cryptocurrency

↓

View Market Data

↓

Historical Price Chart

↓

Analyze Trends

## 7. Logout Flow

User

↓

Logout

↓

Session Ended



Login Page

## **SYSTEM ARCHITECTURE**

### **Frontend**

- React
- HTML5
- CSS3
- Tailwind CSS
- JavaScript (ES6+)

### **Backend**

- Python
- FastAPI
- Uvicorn
- Pydantic

### **Database**

- PostgreSQL (Production)
- SQLite (Development)

### **External APIs**

- CoinGecko API

### **Authentication**

- JWT (JSON Web Tokens)
- bcrypt Password Hashing

## **Charts & Data Visualization**

- Recharts

## **Version Control**

- Git
- GitHub

## **Deployment / Cloud**

- Vercel (Frontend)
- Render (Backend)
- PostgreSQL Cloud Database

## **DATABASE REQUIREMENTS**

### **Entities**

#### **Users**

- user\_id
- full\_name
- email
- password\_hash
- profile\_image (optional)
- created\_at

#### **Cryptocurrencies (API Data - Not Stored Permanently)**

- coin\_id
- name
- symbol
- current\_price
- market\_cap
- volume
- price\_change

- image\_url

## **Watchlist**

- watchlist\_id
- user\_id
- coin\_id
- added\_at

## **Portfolio**

- portfolio\_id
- user\_id
- coin\_id
- quantity
- purchase\_price
- purchase\_date

## **API REQUIREMENTS**

### **Authentication APIs**

#### **Register User**

POST /api/auth/register

#### **Request**

```
{  
  "name": "",  
  "email": "",  
  "password": ""  
}
```

#### **Response**

```
{  
  "success": true,  
}
```

```
  "message": "User registered successfully"
}
```

## **Login**

POST /api/auth/login

## **Logout**

POST /api/auth/logout

## **Cryptocurrency APIs**

### **Get All Cryptocurrencies**

GET /api/coins

### **Get Coin Details**

GET /api/coins/{coin\_id}

### **Search Cryptocurrency**

GET /api/search?q=bitcoin

## **Portfolio APIs**

POST /api/portfolio

GET /api/portfolio

PUT /api/portfolio/{id}

DELETE /api/portfolio/{id}

## **Watchlist APIs**

POST /api/watchlist

GET /api/watchlist

DELETE /api/watchlist/{id}



## **UI/UX REQUIREMENTS**

### **Screens**

#### **Landing Page**

- Hero Section
- Live Market Overview
- Top Trending Coins
- Features Section
- Get Started Button

#### **Login Page**

- Email
- Password
- Remember Me
- Forgot Password

#### **Dashboard**

- Market Statistics
- Top Gainers & Losers
- Search Bar
- Watchlist
- Portfolio Summary

#### **Coin Details Page**

- Current Price
- Market Cap
- Volume
- Historical Price Chart
- Add to Watchlist
- Add to Portfolio

#### **Portfolio Page**

- Holdings
- Profit/Loss
- Portfolio Value

## **Profile Page**

- Edit Profile
- Change Password
- Logout

## **ACCEPTANCE CRITERIA**

### **Feature: User Login**

#### **Given**

A registered user has a valid account.

#### **When**

The user enters valid credentials.

#### **Then**

The user is successfully redirected to the dashboard.

### **Feature: Search Cryptocurrency**

#### **Given**

The user is on the dashboard.

#### **When**

The user searches for a cryptocurrency.

#### **Then**

Relevant search results should be displayed instantly.

## **Feature: Add to Watchlist**

### **Given**

The user is logged in.

### **When**

The user clicks **Add to Watchlist**.

### **Then**

The selected cryptocurrency should appear in the user's watchlist.

## **RISKS & MITIGATION**

| <b>Risk</b>                   | <b>Impact</b> | <b>Mitigation</b>                   |
|-------------------------------|---------------|-------------------------------------|
| <b>Crypto API unavailable</b> | High          | Retry mechanism and cached data     |
| <b>Server Downtime</b>        | High          | Cloud deployment and backups        |
| <b>Data Breach</b>            | High          | JWT, HTTPS, bcrypt encryption       |
| <b>Slow API Response</b>      | Medium        | Response caching and optimization   |
| <b>Invalid User Input</b>     | Medium        | Input validation and error handling |

## **DEPENDENCIES**

### **Internal**

- Development Team
- Project Guide
- UI/UX Design

### **External**

- CoinGecko API
- PostgreSQL Database
- Email Service
- Cloud Hosting Platform

## **MILESTONES & TIMELINE**

| Phase                                      | Duration |
|--------------------------------------------|----------|
| <b>Research &amp; Requirement Analysis</b> | 1 Week   |
| <b>UI/UX Design</b>                        | 1 Week   |
| <b>Database Design</b>                     | 3 Days   |
| <b>Frontend Development</b>                | 2 Weeks  |
| <b>Backend Development</b>                 | 2 Weeks  |
| <b>Integration</b>                         | 1 Week   |
| <b>Testing</b>                             | 1 Week   |
| <b>Deployment</b>                          | 2–3 Days |
| <b>Documentation &amp; Presentation</b>    | 3 Days   |

## **TESTING PLAN**

### **Unit Testing**

Verify individual backend APIs and frontend components.

### **Integration Testing**

Verify communication between frontend, backend, database, and external APIs.

### **System Testing**

Validate complete application functionality from login to portfolio management.

### **User Acceptance Testing (UAT)**

Verify the application meets the project requirements and user expectations.

## **LAUNCH PLAN**

### **Pre-Launch**

- Complete testing
- Finalize documentation
- Prepare deployment environment

### **Launch**

- Deploy frontend

- Deploy backend
- Connect database
- Verify production environment

## **Post-Launch**

- Fix reported bugs
- Optimize performance
- Plan future feature enhancements

## **SUCCESS METRICS**

### **Product Metrics**

- User Registration Rate
- Daily Active Users (DAU)
- Monthly Active Users (MAU)
- User Retention Rate
- Portfolio Creation Rate
- Watchlist Usage

### **Technical Metrics**

- API Response Time
- System Uptime
- Error Rate
- Average Page Load Time

## **APPENDIX**

### **Glossary**

- **Cryptocurrency:** A digital currency secured using cryptographic techniques.
- **Blockchain:** A decentralized digital ledger that records transactions securely.
- **Market Capitalization:** Total market value of a cryptocurrency.
- **Portfolio:** A collection of cryptocurrency investments owned by a user.

- **Watchlist:** A personalized list of cryptocurrencies a user wishes to monitor.
- **JWT:** JSON Web Token used for secure authentication.
- **API:** Application Programming Interface used to exchange data between systems.

## References

- [CoinGecko API Documentation](#)
- [FastAPI Documentation](#)
- [React Documentation](#)
- [PostgreSQL Documentation](#)
- [JWT Documentation](#)

## Change Log

| Version | Date         | Changes                          |
|---------|--------------|----------------------------------|
| 1.0     | 05 July 2026 | Initial Draft                    |
| 1.1     | —            | Feature Enhancements<br>(Future) |